B Tech III Sem 2nd year

|SET| D

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B. Tech II Year, III Semester, First Term Exam-2014

AHE-201: Ethics & Values

Time: 01 ½ Hours
Max. Marks: 20

Note: This paper consists of two passages. Carefully read the passages and answer the following multiple choices questions. Carefully fill all the information in OMR sheet neatly in the blocks. No partial marking.
Full marks (assigned) will be given for a complete answer that means all correct choices.

Section A

Passage-1 (Attempt All Questions: 1X10= 10 Marks)

Ethics offers an explanation to the questions: what is right and wrong, good and bad, and what ought to be done or what ought not to be done? In other words, Ethics provides the guidelines for our conduct, choices, actions, and decisions. Rationality is a key to ethics. It critically provides a certain reasons to explain the right course of choice, action and decisions. In this way rationality is a basis for ethics. What could be the most challenging other than making a decision? Indeed, it's ethical decision making. Decision making is always a tough task to perform but deciding in a situation where the choices are neither completely right nor completely wrong; the decision making gets tougher. In decision making a person is tasted in some pressing situations. The ethical point of view is necessary for a decision maker. It may be the case that someone decides through certain reasons, however, does not think from an ethical point of view whether his/her decision would affect someone or not. This is the reason why an instrumental decision making differs from that

of ethical decision making (hereafter, E.D.M.). In instrumental decision making process though we apply reasoning and critical thinking but the process aims to achieve desired results. The process is primarily concerned with non-ethical result oriented concerns. Cost-benefit analysis and bottom line approaches are some instances of instrumental decision making. In E.D.M. we consider not only the desired results but also the rightness and wrongness of our decisions. E.D.M is not simply a play of rationality and critical thinking but presupposes a moral agency: A value judgment capacity to decide on ethical concerns. Value based ethical concerns in reference to the individual freedom, social welfare, social responsibility, environmental protection and virtues. There are three prominent approaches of ethical decision making: 1) Consequentialism, 2) Deontology, and 3) Virtue.

Another, issue that is rather important in ethical discourse is about justice. What is justice? This question is both procedural and distributive. Justice depends on the principles and processes by which we allocate powers and positions. We distribute resources and other basic life needs. We distribute opportunities, jobs and educational seats amongst the individuals. So, the question of justice is about fair distribution and social equality. The idea is advanced by the great socio-political thinker John Rawls. His view is called justice as fairness: Distributive Justice. On contrary, there is view proposed by great ancient philosopher Aristotle: The Virtue Theory. The view defends the merit that the question of justice is to respect one's talent, merit and skills. The question of justice is teleological and not distributive. According to Aristotle justice is to create a better through meritorious people who can offer best practices through their best skills and talents. However, there is another view of justice propounded by the Individualists: Justice is neither a distributive nor attributive (based on merit and qualities). Justice is to give individuals equal rights and opportunities without any inequality and partiality based on cast, gender, status, background and any such history. Thus, we have three prominent approaches to the idea of justice. These prominent approaches try to defend three different virtues: 1) Social Welfare, 2) Virtues and 3) Individual Freedom.

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1) Which of the following statements are **NOT** correct about the Ethical Decision Making?
   a) It is concerned with non-ethical concerns.
   b) It is necessary for a decision maker.
   c) It is an instrumental sort of decision making.
   d) It's unnecessary for a decision maker.
2) What is **NOT** true about justice?
   a) It's neither procedural nor distributive.
   b) It's only a procedural notion.
   c) It's only a distributive notion.
   d) It is both procedural and distributive notion.
3) Which of the following statements are **NOT** right about distributive justice?
   a) It tries to maintain the view of social equality.
   b) It's justice as fairness.
   c) It examines the fairness of the distribution of resources and jobs.
   d) It's about talent and skills.
4) Which of the following statements are right about ethics?
   a) It is an unethical idea.
   b) It offers an explanation to what is right or wrong.
   c) It provides guideline to our choices, actions and decisions.
   d) It is related to non-ethical concerns.
5) Which of the following statements are **FALSE** about decision making?
   a) Decision making gets tougher when choices are neither completely right not fully wrong.
   b) Decision making is more challenging than ethical decision making.
   c) It requires both rationality and critical thinking.
   d) None of these.
6) What is the value based ethical concerns?
   a) They are concerned with environmental safety and social welfare.
   b) The concern is about social responsibility.
   c) It's only concerned with individual freedom.
   d) They are in reference to individual freedom and virtue.

7) What are the three approaches to justice?
   a) Justice as fairness.
   b) Justice based on merit, talent and skills.
   c) Justice as Unfairness.
   d) Justice based on individual rights.

8) What is **Wrong** about instrumental decision making?
   a) In this process though we apply rationality and critical thinking but it aims to achieve desired results.
   b) The process is based on non-ethical concerns.
   c) Cost-benefit analysis and bottom line approaches are some example of instrumental decision making.
   d) None of these.

9) What is virtue theory?
   a) The question of justice is teleological.
   b) Distribution of resources should be based on merit.
   c) Meritorious people can offer better practices.
   d) It's about social equaliy.

10) Rationality is a key to ethics. What does that statement refers to?
   a) It is a critical device to explain the reasons.
   b) It is not necessary for ethics.
   c) It is a basis for ethics.
   d) All above.

Section B

(Attempt All Questions: 1X10 =10 Marks)

Passage-2

Trolley Problem is a hypothetical case in which we taste one's ethical decision making. Hypothetical cases involve some constructed situations in order to test one's real decision making and problem solving skills. In this case as we all know there are three different versions that actually point out the variations in our decision making with the changes in the conditions. However, the case of trolley car shows that our decision making is not simply based on reasons but also amounts to certain moral emotions which in some sense can be said to be reasonable in so far as

18) What is **NOT** right about the hypothetical case?
   a) It is designed to test real decision making situations
   b) Trolley problem is its example.
   c) They are based on real situation.
   d) All above.
19) What is utilitarian mode of reasoning in this case?
   a) It is to have the better consequences.
   b) It is to reduce the maximum pain.
   c) It is to save the more workers' life.
   d) None of these.
20) Why pushing the fat man in second version is wrong?
   a) We don't have right to take others life.
   b) Killing an innocent is wrong.
   c) The nature of action is objectionable.
   d) All above.

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B. Tech. II Year, III Sem., I Mid-Term Examination, 2014-15
AHM- 201: Mathematics –III

Time:- 1 ½ Hours **Max. Marks:-20**

Section - A

Note: Attempt All Questions. **(1 × 5 = 05 Marks)**

Q.1. Find the divergence of the vector $\vec{V} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k}$ at the point $(2,-1,1)$.

Q.2. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then find *grad r*.

Q.3. Determine the value of $curl(u\vec{a})$, if u is scalar function and $\vec{a}$ is vector function.

Q.4. Reduced the equation $x^3 - 3x^2 + 12x + 16 = 0$ into an equation whose roots are decrease by 1 to the roots of given equation.

Q.5. Find the angle the between surfaces $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 - z = 9$ at the point$(2,-1,2)$.

Section - B

Note: Attempt Any three questions. **(2 × 3 = 06 Marks)**

Q.1. Solve the equation $x^3 + 6x^2 + 9x + 4 = 0$ by Cardon's method.

Q.2. Solve the equation $x^4 - 2x^2 + 8x - 3 = 0$ by Descarte's method.

Q.3. Verify Green's theorem for $\int_C [(xy + y^2)dx + x^2dy]$ where C is bounded by curves $y = x$ and $y = x^2$.

Q.4. Find the values of λ and μ so that the surfaces $\lambda x^2 - \mu yz = (\lambda + 2)x$ and $4x^2y + z^3 = 4$ intersect orthogonally at the point $(1,-1,2)$.

Section - C

Note: Attempt Any three questions. **(3 × 3 = 09 Marks)**

Q.1. Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\,\hat{j}$ taken round the rectangle bounded by the lines $x = \pm a, y = 0, y = b$.

Q.2. Solve the equation $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ by Ferrari's method.

Q.3. Show that the vector $\vec{F} = \frac{\vec{r}}{r^3}$ is irrotational as well as solenoidal, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \sqrt{x^2 + y^2 + z^2}$. Find the scalar potential.

Q.4. Verify Gauss divergence theorem for $\vec{F} = xz\,\hat{i} + y^2\,\hat{j} + zy\hat{k}$ taken over the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

First Term Examination, 2014-15
Course: B. Tech. (CSE), Year: II, Semester:III
Data Structures and Algorithm (CSE 211)

Uni. Roll No:

Time: 1.5 Hrs. **Total**

Marks: 20

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Define Time-space trade off.

II. What do you mean by Data Structure? Differentiate between Linear and Non-linear data structure.

III. What do you mean by Abstract Data Types?

IV. Explain the underflow and overflow conditions (mathematical) on a Queue.

V. Write the structure declaration for a doubly linked list.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. What are the advantages and disadvantages of linked list?

II Write an algorithm to implement a link-list based queue with functions for insertion and deletion in it using its basic terminology.

III Explain Tower of Hanoi problem in detail with 3 disks. Solve it showing all its steps.

IV Let a and b denote the positive integers. Suppose a function Q is defined recursively as follows:

$$Q(a, b) = 0 \quad \text{if } a < b$$

$$Q(a,b) = Q(a-b, b)+1 \quad \text{if } b \leq a$$

Find the value of Q(2,3) and Q(14,3).

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Explain the term Asymptotic Notation. Discuss all the notations in detail.

II. Write functions to perform the following operations on a singly linked list:

a) Insert at End

b) Delete a node from end

c) Search an element in sorted list

III. Differentiate between doubly linked list and circular linked list. Write an algorithm for traversing a circular linked list.

IV. Consider the postfix expression P given below. Evaluate P using the stack based algorithm.

P: 16, 7, 2 ,- , /, 2, 1, 5,+, *, +

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B. Tech. II Year III Semester, First Term Examination, 2014-15

CSE-212: Computer Organization

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section-A

Note: Attempt all Questions **(1*5=5)**

I. What is the value of base b if the decimal roots of equation $x^2-(101)_b x+(210)_b=0$ are 7 and 3?

II. How many check bits are needed to detect single bit error in 2000 bits data word (Show all steps)?

III. Find the 2's Complement of 0.156.

IV. Consider two normalized floating point numbers 0.8526×10^3 and 0.345×10^2 find their product.

V. Consider the IEEE floating point number $(C0600000)_{16}$, Represent it into its decimal floating equivalent number.

Section-B

Note: Attempt any three questions **(2*3=6)**

I. How can we reduce carry propagation delay in binary adder? Justify your answer.

II. Complete the following table.

Decimal	Binary	Octal	Hexa-decimal
111			
	1100.101		
		927	
			FC.1A

III. Design an array multiplier that multiplies two 3-bit numbers. How many AND gates require to design it?

IV. What do you mean by bus transfer? Explain CBS with diagram.

Section-C

Note: Attempt any three questions (3*3=9)

I. Describe block diagram of binary addition-subtraction algorithm for sign-magnitude representation. Drive an algorithm in flowchart form for adding and subtracting for two binary numbers.

II. Show the step-by-step multiplication process of (-5)*(17) by using Booth's Multiplication.

III. List and briefly describe the possible states that define an instruction execution. Which architecture is based on st d program concept?

IV. Consider an 11-bit data word stored in memory is 11011111011. Using Single bit error detection algorithm, determine what check bits would be stored in memory with the data word? Suppose while transmission 10th bit of code word is changed, how receiver identified the error in code word? Justify.

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B. Tech. II Year III Semester, First Term Examination, 2014-15

CSE-213: Discrete Mathematics

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Symbols have their usual meanings.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Describe the following set in Roster form :
S = { x : x is a positive integral divisor of 60 }

II. If a bag contains several white, red, and black shoes then find the minimum number of shoes that a person needs to choose in order to get two pairs (=4) shoes of the same color.

III. Find a simple expression for G.F. of the numeric function
1, -2, 3, -4, 5, -6, ...

IV. Find the number of distinguishable permutations of the letters in the word "BANANA".

V. Find the number of ways that 12 students can be divided into three teams so that each team contains 4 students.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Prove that A U (B - C) = (A U B) - (C - A)

II. Prove that A X (B ∩ C) = (AX B) ∩ (AXC)

III. There are 6 gentlemen and 4 ladies to dine at a round table. In how many ways can they sit themselves so that no two ladies are together?

IV. Find the general solution of the recurrence relation

$$a_r - 4a_{r-1} + 3a_{r-2} = 2^r \text{ , } r \geq 2 \text{ and } a_0=1, a_1=11$$

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Consider a relation R defined on the set of integers as $R = \{(a, b) : a \in Z, b \in Z, a - b \text{ is divisible by } 7\}$. Prove that R is an equivalence relation.

II. Consider the first 1000 positive integers. Out of these integers find the number of integers that are

a. Neither divisible by 2, nor by 3, nor by 5.

b. Number of integers divisible by 2 & 3 only.

III. Solve the following recurrence relation using Generating Functions:

$$a_r - 2a_{r-1} - 3a_{r-2} = 0, n \geq 2. \; a_0 = 3, a_1 = 1$$

IV. A coin is flipped eight times where each flip comes up either heads or tails. How many possible outcomes

a.) are there in total?

b.)contain exactly two heads?

c.)contain the same number of heads and tails?

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B. Tech. II Year III Semester, First Term Examination, 2014-15

ITE 211: Object Oriented Programming

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C
2. Symbols have their usual meanings.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What is Unified Modeling Language?

II. Describe Multiplicity with symbols.

III. What are the command line arguments? Give an example.

IV. Write the name of java's access modifiers and explain what **private** does?

V. Does constructor have a return type? When is a constructor executed?

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. What is modeling? Describe the conceptual model of UML.

II. Write short notes on any two.

(a) Generalization (b) Association (c) Aggregation

III. Explain at least eight salient features of java.

IV. Explain the concept of overloading with the help of suitable example.

Section C

Note: Attempt any Three Questions (3x3=9)

I. What do you mean by constructor? Write a program using Arguments constructor and Copy constructor.

II. Prepare a class diagram for a graphical document editor that supports grouping. Assume that a document consists of several sheets. Each sheet contains drawing objects, including text, geometrical objects, and groups. A group is simply a set of drawing objects, possibly including other groups. A group must contain at least two drawing objects. A drawing object can be a direct member of at most one group. Geometrical objects include circles, ellipses, rectangles, lines and squares.

III. Define Events and States. Draw a state diagram for telephone line.

IV. What do you mean by Actors and Use Cases? Design a Use Case Diagram for a vending machine.